$$W = \int_{V_i}^{V_f} P dV \qquad (8.40)$$

By assuming that the adiabatic process occurs quasi-statically, at every stage the ideal gas law is valid. Under this condition, the adiabatic equation of state is PV^{γ} = constant (or)

P = $\frac{\text{constant}}{V^{\gamma}}$ can be substituted in the equation (8.40), we get

$$\therefore W_{adia} = \int_{v_i}^{V_f} \frac{\text{constant}}{V^{\gamma}} dV$$

$$= \text{constant} \int_{V_i}^{Vf} V^{-\gamma} dV$$

$$= \text{constant} \left[\frac{V^{-\gamma+1}}{-\gamma+1} \right]_{V_i}^{V_f}$$

$$= \frac{\text{constant}}{1-\gamma} \left[\frac{1}{V_f^{\gamma-1}} - \frac{1}{V_i^{\gamma-1}} \right]$$

$$= \frac{1}{1-\gamma} \left[\frac{\text{constant}}{V_f^{\gamma-1}} - \frac{\text{constant}}{V_i^{\gamma-1}} \right]$$

But, $P_i V_i^{\gamma} = P_f V_f^{\gamma} = constant.$

$$\therefore W_{adia} = \frac{1}{1-\gamma} \left[\frac{P_f V_f^{\gamma}}{V_f^{\gamma-1}} - \frac{P_i V_i^{\gamma}}{V_i^{\gamma-1}} \right]$$

$$W_{adia} = \frac{1}{1-\gamma} [P_f V_f - P_i V_i] \qquad (8.41)$$

From ideal gas law,

$$P_f V_f = \mu R T_f \text{ and } P_i V_i = \mu R T_i$$

Substituting in equation (8.41), we get

$$\therefore W_{adia} = \frac{\mu R}{\gamma - 1} [T_i - T_f] \qquad (8.42)$$

In adiabatic expansion, work is done by the gas. i.e., W_{adia} is positive. As $T_i > T_f$, the gas cools during adiabatic expansion.

In adiabatic compression, work is done on the gas. i.e., W_{adia} is negative. As $T_i < T_f$, the temperature of the gas increases during adiabatic compression.

Note

Even though we have derived equations (8.41) and (8.42) by assuming that the adiabatic process is quasi-static, both the equations are valid even if the process is not quasi-static. This is because P and V are state variables and are independent of how the state is arrived.

In the adiabatic PV diagram shown in the Figure 8.32, the area under the adiabatic curve from initial state to final state will give the total work done in adiabatic process.

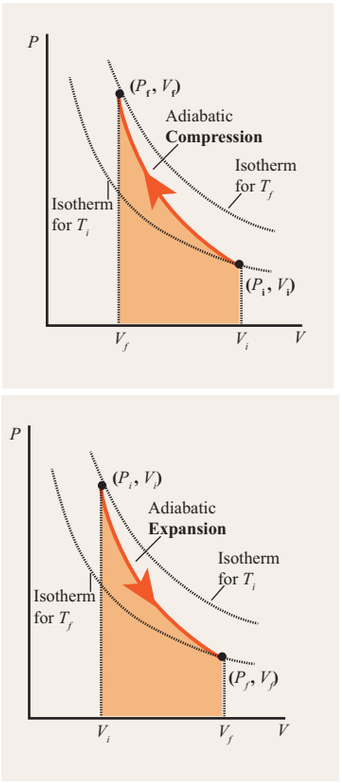


Figure 8.32 PV diagram -Work done in the adiabatic process

To differentiate between isothermal and adiabatic curves in (Figure 8.32) the adiabatic curve is drawn along with isothermal curve for T_f and T_i. Note that adiabatic curve is steeper than isothermal curve. This is because $\gamma > 1$ always.

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